

Tacit: Confidential Assets and DeFi on Bitcoin, Bridged by Zero-Knowledge Reflection

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Abstract

Bitcoin holds value without a trusted party, but issuing assets on it, hiding amounts or trading them has meant accepting one: an off-chain proof courier, a federation, a rollup operator or a custodian. Tacit removes that party.

On Bitcoin, Tacit is an indexer-validated metaprotocol. Assets live in Taproot envelopes, amounts are Pedersen commitments with range proofs, and conservation is a Schnorr signature over the excess. Any indexer running the specification reaches the same state from the chain alone.

Tacit reflects that state into an immutable confidential pool on Ethereum, where every settlement is one SP1 zero-knowledge proof. Bitcoin reaches the pool through a proof-of-work header relay and a guest that re-verifies every Tacit envelope. The pool reaches Bitcoin through a light-client proof verified inside that same guest. No signer stands between the chains.

The pool runs confidential DeFi: private transfers, swaps and routes, OTC trades and bids, atomic swaps, stealth payments, a CDP issuing cUSD, farms, and cBTC, minted against real BTC in self-custody locks. Any op can be relayed without gas, with the fee bound inside the proof, and every balance recovers from a key. The core never changes. The protocol grows at its edges and through a lineage of pools, each deploying its successor, toward a V2 built on Bitcoin covenants.

1 Introduction

A useful asset layer on Bitcoin needs five properties:

1. its state is public data anyone can check;
2. amounts are hidden;
3. value can move to where computation is cheap without trusting a bridge;
4. BTC itself can back a fungible asset;
5. a user can recover everything from a key.

Existing designs each give up one:

Design	What it gives up
Ordinals, Runes	Amounts are public.
RGB, Taproot Assets	Validity moves off-chain, so losing a proof means losing the balance.
Liquid	Relies on a federation.
Rollups and Bitcoin L2s	Rely on an operator set.
Wrapped BTC	Relies on a custodian or a threshold group.

Tacit provides all five by composing existing systems rather than adding consensus. Bitcoin orders the data and indexers read it (§2). Commitments hide amounts (§3). An immutable contract on Ethereum accepts only state transitions that carry a zero-knowledge proof (§4). The chains learn about each other through proofs, not signatures (§5). Real BTC backs cBTC through self-custody locks, secured economically until Bitcoin gains covenants (§6, §11). Every balance recovers from a key (§7).

2 A metaprotocol on Bitcoin

A Tacit op is a commit/reveal pair. The commit pays to a Taproot output whose only script leaf is:

$$\langle P \rangle \text{ OP_CHECKSIG OP_FALSE OP_IF "TACIT" 0x01 } \langle \text{payload} \rangle \text{ OP_ENDIF}$$

The reveal spends the commit by script path, so the payload rides in witness data. Bitcoin nodes do not interpret it. Indexers do: a Tacit UTXO is valid iff the op that created it passes its rules and each of its asset inputs is itself valid.

The rules are deterministic, so two indexers that see the same bytes reach the same verdict. There is no quorum, no vote and no leader, the same trust model as Ordinals and Runes. Users trust the published specification and its open reference implementations. The Bitcoin reflection guest (§5) is a second, independent check: it re-implements the rules for every op it folds, under a verifying key that cannot change.

Ops cover:

- **Issuance:** confidential supply, or fair-launch mints against a public cap. TAC, the native asset, was issued with a zero mint authority, so its supply is fixed at 21 million.
- **Transfers:** hidden amounts, with optional stealth addresses.
- **Trades:** atomic trades against BTC, bids that fill while the buyer is offline, and a native AMM with swaps, routes, uniform-price batches and LP farms.
- **cBTC:** locks and redeems of real BTC.
- **Cross-chain:** bridge burns, re-mints of notes returning from Ethereum, and value-free calls in both directions.

An unknown opcode creates no state, so the protocol grows by adding opcodes.

3 Confidential value

Every amount is a Pedersen commitment on secp256k1:

$$C = v \cdot H + r \cdot G,$$

where G is Bitcoin's generator and H is a nothing-up-my-sleeve point hashed by SHA-256 from the tag `tacit-generator-H-v1`. Commitments hide the value perfectly and add homomorphically. A transaction balances iff its excess

$$E = \sum C_{\text{out}} + b \cdot H - \sum C_{\text{in}}$$

carries no H component, where b is any public burn. Tacit proves this with a BIP-340 signature under E , a Mimblewimble-style kernel. Only someone who knows every blinding can sign, and nobody can sign for an excess that contains value.

Range proofs bound every output to $[0, 2^{64})$, aggregated over up to eight outputs. New transfers use Bulletproofs+, and classic Bulletproofs remain valid. The recipient's blinding and an 8-byte amount keystream come from an ECDH secret between sender and recipient, anchored to the transaction's first asset input. A recipient finds and opens every credit from its own key and the chain, with no share link and no sync server.

The same commitment, with the same G and H , is the note in the Ethereum pool. One note format on both chains lets value cross between them without changing representation.

4 The confidential pool

The pool is one immutable contract with no owner and no pause. It holds three depth-32 keccak Merkle trees, for notes, locks and CDP positions, each with its own spent set. It also escrows ETH and outside tokens, and holds the reflected state of Bitcoin.

A note is $(asset, C, owner)$. A pool-native note's leaf is a keccak hash of those fields. A Bitcoin-homed note's leaf commits instead to the note's Bitcoin key and the deployment it is bound to. Nullifiers derive from the leaf. An owned note's nullifier also takes the owner's secret nullifier key, so an observer cannot tell which leaf a spend consumes. Bearer and Bitcoin-homed notes use a nullifier derived from the leaf alone.

Entry and exit. `wrap` escrows ETH or any ERC-20 and records a deposit that only its depositor's proof can consume. One transaction can wrap and immediately spend the deposit as a transfer, swap, liquidity add or CDP mint. Assets the pool mints itself (TAC, cBTC, cUSD) need no escrow: unwrapping mints the canonical ERC-20, and wrapping it back burns it. A note can also exit partway, as a public payout plus hidden change.

Settlement. Every change to notes, locks and positions is a batch of up to 256 ops proven by one SP1 program. For each op, the program checks:

- that each input leaf is a member of a known root, without revealing which leaf;
- the range proofs;
- the kernel or opening proofs;
- the op's own rules.

It then commits the public effects: nullifiers, new leaves, payouts, fees, swaps, liquidity changes, bridge claims and memos. The contract verifies the proof against a pinned verifying key and applies exactly those effects.

A spend proves membership against any root the tree has ever had. A proof never goes stale, and the anonymity set of an owned-note spend is every owned note of its asset. Nothing accumulates cost with use:

- an insert is 32 hashes at any fill;
- nullifier and root sets are constant-time maps that are never iterated;
- the guest touches history only through fixed-depth proofs.

The contract never records more spent native notes than it has ever created, a floor against inflation that it enforces independently of the proof. Every proof is bound to `CHAIN_BINDING = keccak(chainid||address)`, so a proof for one deployment is invalid in any other.

5 Reflection: a bridge without signers

Bitcoin to Ethereum. Anyone may extend `BitcoinLightRelay`, a header relay that checks full proof-of-work and follows the heaviest chain. The reflection guest proves a run of blocks on top of it. For each block it:

1. re-hashes every transaction to the header’s Merkle root;
2. nullifies every input that spends a reflected note;
3. re-verifies and folds each Tacit envelope it understands, and skips the rest;
4. extends a digest chain over the result.

The pool accepts an attestation only if its prior digest matches, its tip is buried at least 24 blocks under the relay tip, and its counters match the pool’s own. The pool keeps every attested root of Bitcoin’s Tacit notes, together with the current roots of Bitcoin’s spent set and bridge burns, cBTC lock records and pending Bitcoin calls.

Value enters the pool from Bitcoin in two ways:

- **Bridge burn.** The note is destroyed on Bitcoin under a burn id that commits to the target deployment, and the pool mints it exactly once.
- **Fast lane.** A Bitcoin note bound to this deployment is spent directly on Ethereum. Its proof shows that the note’s nullifier is absent from the current reflected Bitcoin spent set, and it carries a signature from the note’s Bitcoin key. The pool records the spend, and reflection retires the note on Bitcoin.

Ethereum to Bitcoin. A second guest, built on the `sp1-helios` sync-committee light client, proves finalized pool storage: cross-out commitments, consumed sources and outbox messages. The committee chains from period to period, with each step committed. The Bitcoin reflection guest verifies this proof recursively and folds, by membership:

- re-mints on Bitcoin of notes crossed out of the pool;
- outbox messages;
- retirements of fast-lane notes.

Calls. A value-free Bitcoin envelope can authorize an Ethereum call, which an executor contract runs once it is reflected. Ethereum can post messages that reflection delivers to Bitcoin.

The bridge’s soundness rests on Bitcoin proof-of-work, the Ethereum sync committee and SP1. There is no multisig, attestor set or IOU. Liveness needs someone to run the provers. Anyone may, and Ethereum-homed notes stay spendable if nobody does. A full round trip has settled on mainnet: a note crossed out of the pool, was re-minted on Bitcoin, was burned there and was minted back, proven in both directions.

6 Confidential DeFi

Swaps. Pools are constant-product curves with a fee tier and an optional protocol fee. Reserves are public. A plaintext AMM contract, `TacitPublicAmm`, trades against the same reserves, so public

and confidential liquidity share one curve.

- **OP_SWAP_ROUTE** swaps a hidden input note through up to four pools, with an end-to-end minimum output.
- **OP_SWAP** clears many hidden-output intents at one uniform price. Traders in one batch pay the same price, so they cannot sandwich each other, and the batch reveals only its net change to reserves.

Prover-blind swaps. Whoever proves an ordinary swap sees its amounts. **OP_SWAP_BLIND** keeps them out of the SP1 witness entirely. Each trader commits to its input on BabyJubJub, the curve native to BN254, and a 169-byte cross-curve sigma proves that commitment equals the trader's secp256k1 note. The batch carries one Groth16 proof of the **amm_swap_batch** circuit, which proves over hidden amounts that the clearing price, every fill and every minimum output are correct. Only the batcher that builds that proof sees the amounts; the SP1 prover and the chain never do. The op is live in the guest, and swaps settle through **OP_SWAP** and **OP_SWAP_ROUTE** until the relay's batch coordinator ships.

Uniform-price clearing over hidden amounts is cheap as a circuit and expensive as a zkVM program, so this one step uses a circuit. Its verifying key comes from the AMM trusted-setup ceremony (Hermez Phase 1, then a Phase 2 of 5,018 contributions sealed by a Bitcoin-block beacon) and is compiled into both SP1 guests. The guests' own verifying keys therefore fix it. The reflection guest verifies the same circuit for **T_SWAP_BATCH** on Bitcoin.

Trades without a pool. **OP_OTC** swaps two parties' notes directly. **OP_BID** is a limit order with a pre-signed grid of partial fills, so the buyer need not be online. Adaptor locks tie a claim to the revelation of a Schnorr signature scalar, which gives atomic swaps against the other chain.

Stealth payments. A payer locks a note under a one-time key derived from the recipient's published key. The recipient claims it with a signature, and the payer may refund after a deadline. A Bitcoin bridge burn can mint straight into such a lock.

cBTC. A user locks BTC on Bitcoin with a **T_CBTC_LOCK** envelope that pre-commits to the note it will mint. Reflection records the lock's value, and the pool mints cBTC once per lock. The locker keeps the lock's key. Anyone may fund a wstETH escrow of at least $1.5\times$ the lock's value, which makes spending the lock unprofitable: a minted lock spent without a matching redeem is visible to reflection and slashable, and each funder reclaims its share after a proven redeem. The amount minted is oracle-free. A price feed only sizes the escrow, so a stale feed can pause a mint but never inflate one. cBTC is live on mainnet as a pilot. Its custody is economic until covenants make it enforced (§11).

cBTC is a Tacit asset on both chains. A cBTC note crosses out of the pool and is re-minted on Bitcoin by light-client proof (§5), where it moves with hidden amounts like any other Tacit note.

cUSD. A CDP locks a basket of cBTC notes and mints cUSD. The position's owner is hidden; its amounts are public, because the controller prices them against a BTC/USD feed. Minting requires 150% collateral today. Below 130% a position can be liquidated: the liquidator repays the debt and takes the basket. A stability fee and a savings rate are built in and dormant. Policy lives in a governed **CollateralEngine** that the pool only ever asks to accept or reject, with on-chain caps on its ratios and fee, and a cooling-off period before a feed change or a stricter liquidation ratio takes effect.

Farms. LP-share notes bond into a receipt note. A farm controller tracks reward-per-share at

entry, pays harvests from a treasury the pool escrows for it, and returns the shares on unbonding. Its governor can reweight pools within capped, delayed bounds, but can never shorten a running stream or lower its total rate.

7 Gasless, self-sovereign, recoverable

Gasless. Any op may be relayed. The relayer’s fee is part of what the proof commits to, bound by the conservation kernel, the opening proof or the note’s spend signature. The relayer pays gas and collects exactly that fee. It cannot redirect a payout, raise the fee or submit after a deadline the user sets. Once value is in the pool, a user never needs ETH to move it. Entering takes one Ethereum transaction, which can also tip a relayer to prove and settle the deposit.

Self-sovereign. The relayer is optional. `settle` is open to anyone, proofs can be generated locally or on an open prover network, and all three guest programs rebuild byte for byte from source. A user can prove locally, settle directly, and involve no third party.

Recoverable. A wallet rebuilds every balance and position from its key and public chain data. Bitcoin credits are found by trial derivation against each input’s sender key (§3). Pool notes carry memos encrypted to their owner and committed in the proof that created them, and positions are read back from the pool’s events. There is no backup file to lose and no server to ask.

8 An immutable core that keeps improving

The pool, its guests and their verifying keys cannot be upgraded, so nobody can change the rules under a note that already exists. The protocol still grows, in two ways that never touch the core.

At the edges. New Bitcoin opcodes are additive, so indexers adopt them without disturbing old ones. Anyone can register an ERC-20 into the pool. New debt assets and reward programs plug in as controllers that the pool consults only to accept or reject. Clients, relayers and provers improve freely, because each only produces proofs the fixed guests already verify. V1 gains depth and features as its market grows, with no migration at all.

By succession. The pool live today is the root of the V1 lineage. Each pool deploys its own successor through `createNextGen`, its one privileged call, which the ops multisig holds. The call runs once, from the pool’s own address, and a successor accepts a predecessor only if that predecessor deployed it. The steward chooses the successor’s code and nothing else: it cannot touch escrow, freeze an exit or redirect a payout. Succession carries state forward:

- **The Bitcoin lane continues without a gap.** The successor’s first reflection proof rebases on the predecessor’s attested digest, tip and counters, read live from its contract. Value held on Bitcoin never has to move.
- **Assets keep their identity.** Asset ids are shared across chains and generations, so TAC, tETH and every Bitcoin asset are the same asset in each pool.
- **The old pool keeps its promises.** It refuses new value, but every exit, every release of value already committed, and its reflection stay open.
- **Users move on their own schedule,** by exiting one pool and entering the next. Nothing moves value on their behalf, and a pool deployed by anyone else is an isolated system that cannot touch the lineage.

9 Privacy

	Hidden	Public
Bitcoin	amounts; stealth recipients; trade sizes in T_SWAP_BATCH	addresses, the transaction graph, asset ids, burn amounts, AMM reserves
Pool	amounts; which owned note a spend consumes	the deposit and withdrawal boundary, AMM reserves and their changes, CDP amounts
Relay	your key, seed and blindings	the witness of the ops it proves

The pool's anonymity set is every owned note of an asset, not one denomination. Proving locally keeps a witness on one device end to end, and OP_SWAP_BLIND keeps trade sizes from the SP1 prover once its coordinator ships.

10 Trust and limits

Relied on:

- Bitcoin and Ethereum consensus;
- the soundness of SP1 and Groth16;
- the sp1-helios sync committee;
- for the swap circuit only, at least one honest ceremony contributor;
- the CollateralEngine price feeds: BTC/USD for cUSD, and BTC per wstETH for cBTC escrow;
- the ops multisig (2-of-4, with a one-hour delay unless all four sign), within the on-chain bounds of the collateral and farm controllers;
- for a walk-away bid on Bitcoin only, a hosted watchtower holding a capped, dedicated key over the sats the buyer commits.

Not relied on: relayers, the hosted API, gateways and explorers. The reference indexer reads Bitcoin from public endpoints, and any other source gives the same state.

Limits:

- Reflection halts, rather than rewrites, on a Bitcoin reorg deeper than its confirmation depth.
- Anyone can submit a relayed proof first and collect its fee. The user's outcome is the same, because the proof works for any relayer.
- cBTC is economically secured until covenants.
- Bitcoin transfers carry a large witness, so Tacit on Bitcoin suits settlement, and the pool is where frequent activity belongs.

11 Covenants, V2 and the road ahead

Bitcoin cannot yet restrict where an output may be spent. Once it can, through CTV, OP_CAT, OP_CHECKSIGFROMSTACK, OP_VAULT or an equivalent, three constructions become fully trustless. Tacit reserves room for each today:

- **Covenant cBTC.** A lock whose sats can leave only through a redemption. Pool op 5, OP_COVENANT_MINT, is held for its mint, which reuses the existing value binding with no escrow.
- **On-chain bid escrow.** A buyer's pre-signed input bound to its bid template, replacing the watchtower and activating the reserved batch-fill and both-sides match bytes.
- **Fractional slots.** Reserved opcodes that split a BTC slot into fungible shares and recombine them, with no shared vault.

V2. V1 is built to hand off to this covenant generation. Its Bitcoin opcodes arrive additively, and its settle logic uses the op codes V1 keeps free. V2 deploys through V1's own `createNextGen` and inherits the Bitcoin lane (§8). cBTC then moves from economic to enforced custody one lock at a time: a locker redeems a V1 lock, which releases its escrow, and re-locks the sats in a covenant vault that mints in V2 with no escrow at all. Every V1 note keeps its exits throughout.

A shielded pool on Bitcoin. Bitcoin opcodes 0x6C and 0x6D are reserved for a shielded pool that lives on Bitcoin alone, described in the companion paper *Secret Sats*. It holds Tacit assets, not BTC, so it adds no custody. Shields move notes in, and exits return value to ordinary Tacit notes. Each spend proves membership, nullifiers, its owner's signature and conservation in one SP1 proof, so sender and receiver are unlinked and amounts stay hidden. Indexers verify each proof locally with no Ethereum dependency, and relayers post spends, so a sender needs no Bitcoin wallet. cBTC is its path to BTC: sats locked on Bitcoin mint cBTC in the Ethereum pool, cBTC crosses back to Bitcoin by light-client proof, and there it shields. The pool runs end to end on signet today. Sats will also move in and out by atomic swap, and by covenant once Bitcoin allows one.

Cryptography evolves the same way: a better range proof is added alongside the old one, as Bulletproofs+ was, and proof systems verifiable in Bitcoin script could move parts of indexer validation into consensus.

12 Conclusion

Tacit is an asset layer on Bitcoin with no bridge signer and no custodian:

- a deterministic metaprotocol with hidden amounts;
- an immutable, proof-gated pool that carries the same notes into confidential DeFi;
- a bridge made of proofs in both directions;
- cBTC, which carries real BTC into that DeFi and back to Bitcoin;
- a lineage of immutable pools that improves as its market grows and hands off to V2 when Bitcoin gains covenants.

Where cryptography can enforce a property, it does. Where Bitcoin cannot yet enforce one, the gap is stated, bounded, and reserved for a covenant to close.

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